

Spoken Tutorial on Understanding APIs

Module	Module 3: An introduction to Building
Episode	2. Understanding APIs
Learning Objective	At the end of this tutorial learner will be able to 1. Define what an API is. 2. Explain the role of an API in AI applications. 3. Identify how API keys secure access. 4. Describe the flow of requests and responses via APIs.
Approx. Duration	3-4 min
Outline	• What an API is and why it matters • Analogy: API as a connector between requests and models • Prompts as inputs sent through APIs • API keys for authentication and access control • How applications send requests and receive AI-generated responses
Meta Tags	API, AI, Python, API keys, authentication, requests, responses, prompts, connector, web services
Pre-requisite Tutorial	Episode 1: Python basics for API

Script

Visual Cue	Narration
Title Slide	Welcome to this Spoken Tutorial on Understanding APIs .
Learning Objectives Slide	In this tutorial, you will learn to, • Define what an API is. • Explain the role of an API in AI applications. • Identify how API keys secure access. • Describe the flow of requests and responses via APIs.
System Requirements Slide	Here I am using a browser on a computer or mobile.
Pre-requisite Slide	To follow this tutorial, you should be familiar with Python basics. These basics are covered in Episode 1: Python basics for API .

Illustration showing API as a waiter connecting a customer (application) to a kitchen (AI model)	Have you ever wondered how different software applications talk to each other? An API, or Application Programming Interface, is that digital connector. It allows various software applications to communicate seamlessly. Think of an API like a waiter in a restaurant. You give your order, or request, to the waiter. The waiter takes your order to the kitchen, which is like the AI model. Then, the waiter brings back your meal, or the response. That's how an API works to connect different parts.
Illustration showing a prompt being sent through an API to an AI model	Now, how do we tell an AI model what we want it to do? In AI applications, we send prompts through an API. These prompts are specific instructions or data. The AI model then processes these prompts. It generates the desired output based on your instructions. So, an API is the bridge for your prompts to reach the AI.
Illustration showing a key unlocking access to an API	But how do APIs ensure that only authorized users or applications can access them? This is where API keys come in. API keys are unique identifiers. They are used for authentication and access control. They ensure that only authorized applications can interact with the API. This protects both the service and your valuable user data.
Illustration showing the flow from application sending a request, through API, to AI model, and back with a response	Let's put it all together and see how applications use APIs. First, your application sends a request to an API. The API then relays this request to an AI model for processing. Once the AI model generates a response, the API takes it. It then delivers the response back to your application. This completes the entire communication cycle smoothly.
Summary Slide	Let us summarize what we learned. We defined APIs as digital connectors. We explored how prompts are sent through them. We understood the role of API keys for security. Finally, we traced the request-response flow in AI applications.
Assignment Slide	Now as an assignment, Explore the documentation of a public API, like a weather API. Identify its key endpoints and required authentication methods. Compare the results.

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